PUNE INSTITUTE OF COMPUTER TECHNOLOGY

DHANKAWADI, PUNE –43

Class: B.E. **Semester:** I

List of Assignments

Subject: Laboratory Practice IV (410247)

EXP. NO	PROBLEM STATEMENT
410244 (C): Cyber Security and Digital Forensics	
Group 1	
1.	Write a program for Tracking Emails and Investigating Email Crimes. i.e. Write a program to analyse e–mail header.
2.	Implement a program to generate and verify CAPTCHA image.
3.	A person on a nearby road is trying to enter a WiFi network by trying to crack the password to use the IP Printer resource; write a program to detect such an attempt and prohibit the access. Develop the necessary scenario by Using an IEEE 802.11, configure a Wi-Fi adapter and Access Point.
4.	Write a computer forensic application program for Recovering permanent Deleted Files and Deleted Partitions.
5.	Write a program for Log Capturing and Event Correlation.
Group 2 **(01 Mini- project)**	
1.	**Mini–project-** Design and develop a tool for digital forensic of images.
2.	**Mini Project -** Design and develop a tool for digital forensic of audio.
3.	**Mini Project -:** Design and develop a tool for digital forensic of video.
4.	**Mini Project** - Design a system for the analysis of cyber-crime using various cyber forensic techniques and compare each technique with respect to integrity, confidentiality, availability.

410245(A): Information Retrieval	
Group 1	
1.	Write a program to Compute Similarity between two text documents.
2.	Implement Page Rank Algorithm.
3.	Write a map-reduce program to count the number of occurrences of each alphabetic character in the given dataset. The count for each letter should be case-insensitive (i.e., include both upper-case and lower-case versions of the letter; Ignore non-alphabetic characters).
4.	Write a program to implement a simple web crawler.
5.	Write a program to parse XML text, generate Web graph and compute topic specific page
Group 2 **(01 Mini- project)**	
1.	Mini project: Develop Document summarization system
2.	Mini Project: Develop Tweet sentiment analysis system
3.	Mini Project: Develop Fake news detection system
4.	Mini Project: Develop a Abstractive summarization system

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